

LP-Diff: Towards Improved Restoration of Real-World Degraded License Plate

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Abstract

License plate (LP) recognition is crucial in intelligent traffic management systems. However, factors such as long distances and poor camera quality often lead to severe degradation of captured LP images, posing challenges to accurate recognition. The design of License Plate Image Restoration (LPIR) methods frequently relies on synthetic degraded data, which limits their effectiveness on real-world severely degraded LP images. To address this issue, we introduce the first paired LPIR dataset collected in real-world scenarios, named MDLP, including 10,245 pairs of multi-frame severely degraded LP images and their corresponding clear images. To better restore severely degraded LP, we propose a novel Diffusion-based network, called LP-Diff, to tackle real-world LPIR tasks. Our approach incorporates (1) an Inter-frame Cross Attention Module to fuse temporal information across multiple frames, (2) a Texture Enhancement Module to restore texture information in degraded images, and (3) a Dual-Pathway Fusion Module to select effective features from both channel and spatial dimensions. Extensive experiments demonstrate the reliability of our dataset for model training and evaluation. Our proposed LP-Diff consistently outperforms other state-of-the-art image restoration methods on real-world LPIR tasks. The dataset and code are available at <https://github.com/haoyGONG/LP-Diff>.

1. Introduction

License plate (LP) recognition has emerged as a significant research area in recent years. However, in real-world scenarios such as traffic crime investigations, security surveillance, and intelligent transportation systems, captured LP images often suffer from severe degradation due to environmental factors, distance, and camera quality. These degradations result in low resolution, distortion, and poor contrast, which can lead to substantial errors in existing auto-



Figure 1. Visual comparison of our model LP-Diff with other state-of-the-art models on real-world LPIR tasks. The text sequence above each image is the identification result from the LP recognition model. Red characters indicate recognition errors at those positions, while a red '-' indicates an unrecognizable character at this position.

mated recognition systems. Consequently, restoring low-resolution LP images is a practical and challenging task, aimed at mitigating degradation, enhancing resolution, and ultimately restoring the images to a recognizable state.

In the general domain, various deep-learning methods have been proposed to address image restoration problems [10, 13, 14, 37, 45, 48]. Based on these methods, the first LPIR work [42] was proposed, followed by MSR-CNN [41], which used an architecture similar to Inception [28] to capture various features of low-resolution LP images. DP-GAN [21] utilized complementary information to restore LP images, leveraging the Generative Adversarial Network (GAN) [8]. However, these methods perform well in restoring synthetic degradations but fail to achieve satisfactory accuracy when handling real-world complex degradations. To justify it, we evaluate the performance of several state-of-the-art (SOTA) image restoration algorithms. As shown in Figure 1, we can conclude that all these SOTA methods fail to perform well in real-world tasks.

Along with the development of LPIR models, several LP datasets [12, 17, 19, 22, 40] have been introduced. Nonetheless, these datasets primarily consist of unpaired high-resolution single-frame images, which need to be artificially degraded using synthetic degradation models [22]

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to generate low-resolution images for model training. The synthetic images exhibit less degradation and consistency compared to real-world images. Accordingly, existing image restoration methods rarely tackle severe degradations, making them ineffective for practical LPIR tasks. We conduct statistical and visual comparisons in Section 4.6 to validate it.

To address these challenges, we propose a comprehensive solution consisting of a real-world multi-frame degraded LP dataset and a dedicated LPIR model. The dataset, named MDLP, includes 10,245 data groups, each containing multiple consecutive frames of degraded LP images alongside their corresponding high-resolution images, all automatically captured using high-speed cameras in real road environments. For the model, we aim to propose a new framework based on generative models to better cooperate with our dataset MDLP. However, the commonly used GAN models are unstable during training and prone to mode collapse. Inspired by the powerful generative capabilities of the recently proposed Diffusion model [11], we propose a novel Diffusion-based end-to-end model, LP-Diff, leveraging texture and temporal information from consecutive frames to achieve real-world LP restoration that surpasses existing SOTA models. Specifically, we use an encoder and the Inter-frame Cross Attention Module (ICAM) to integrate information from three consecutive frames of the degraded images. A crucial aspect of this stage is exploiting temporal information, enabling the model to perceive the relational changes between frames. Next, we design the Texture Enhancement Module (TEM) to compute curvature and gradient magnitude to enhance the texture information. Additionally, to effectively filter features across both spatial and channel, we propose the Dual-Pathway Fusion Module (DFM), which applies weighting separately on spatial and channel dimensions. Finally, we calculate the residual from the fused result to guide the Diffusion model at each time step, ultimately producing a clear LP image.

In summary, our main contributions are as follows:

- We propose LP-Diff, a Diffusion-based end-to-end model for significant LP image restoration. Our model includes an Inter-frame Cross Attention Module (ICAM) and a Dual-Pathway Fusion Module (DFM), which effectively fuse temporal dependencies and texture information.
- We propose the Texture Enhancement Module (TEM), which strategically utilizes curvature and gradient magnitude to restore the geometric shape and edge intensity of texture.
- We introduce a large-scale multi-frame degraded LP dataset, named MDLP, which is fully collected from real-world scenarios. The dataset comprises 10,245 meticulously selected data groups, each containing multiple consecutive frames of degraded LP images along with their corresponding clear images.
- Experimental results show that our proposed LP-Diff outperforms previous image restoration methods in the real world, and extensive ablation studies validate the effectiveness of our contributions.

2. Related Work

2.1. Image Restoration

Image restoration is a crucial research topic in computer vision. After the introduction of the pioneering CNN-based super-resolution work SRCNN [5] and denoising work DnCNN [46] introduced a multi-layer convolutional neural network to map the correlations between low-resolution and high-resolution images, image restoration has received widespread attention [20, 24, 38, 39, 44, 49]. For example, DFF [48] employed dense feature fusion to create a deeper network with higher training efficiency and stronger restoration capabilities. Uformer [33], based on transformer architecture, introduced a hierarchical encoder-decoder and a restoration modulator, enhancing the ability to capture both local and global dependencies in image restoration. Mambair [10] improved on Mamba by incorporating local enhancement and channel attention to address local pixel forgetting and channel redundancy issues. After the introduction of the Diffusion model [11], numerous Diffusion-based methods [4, 6, 23, 36, 43, 51] have been applied to image restoration. However, many of these methods are based on specific degradation assumptions and lack generalization to real-world degradations [9]. Therefore, some approaches [30, 31, 35] use degradation models to synthesize real-world-like degradations but rely on a single model to handle multiple degradations. In this paper, the proposed method not only uses a single model to address complex degradations but also utilizes datasets collected from real-world scenarios rather than synthetic data.

2.2. License Plate Image Restoration

LPIR was initially proposed [42] to improve recognition accuracy, followed by some methods [7, 50] based on traditional image processing. With advancements in deep learning and GAN [8] methods, CNN-based [22, 41] and GAN-based [18, 19, 21] LPIR methods have been developed. After the Diffusion model [11] applied in image generation, DiffPlate [2] employed a CNN-based conditioning module to guide the Diffusion model in denoising LP images. Alongside the development of LPIR models, datasets for LP have also emerged. CCPD [40] is a dataset of Chinese city parking lot LP, containing images with different viewing angles or occlusions. ALOP [12] and UFPR-ALPR [17] also include thousands of single images of LP from different countries. However, most of these datasets only contain high-resolution images, leading to the proposal of degradation models [22] to generate low-resolution images from

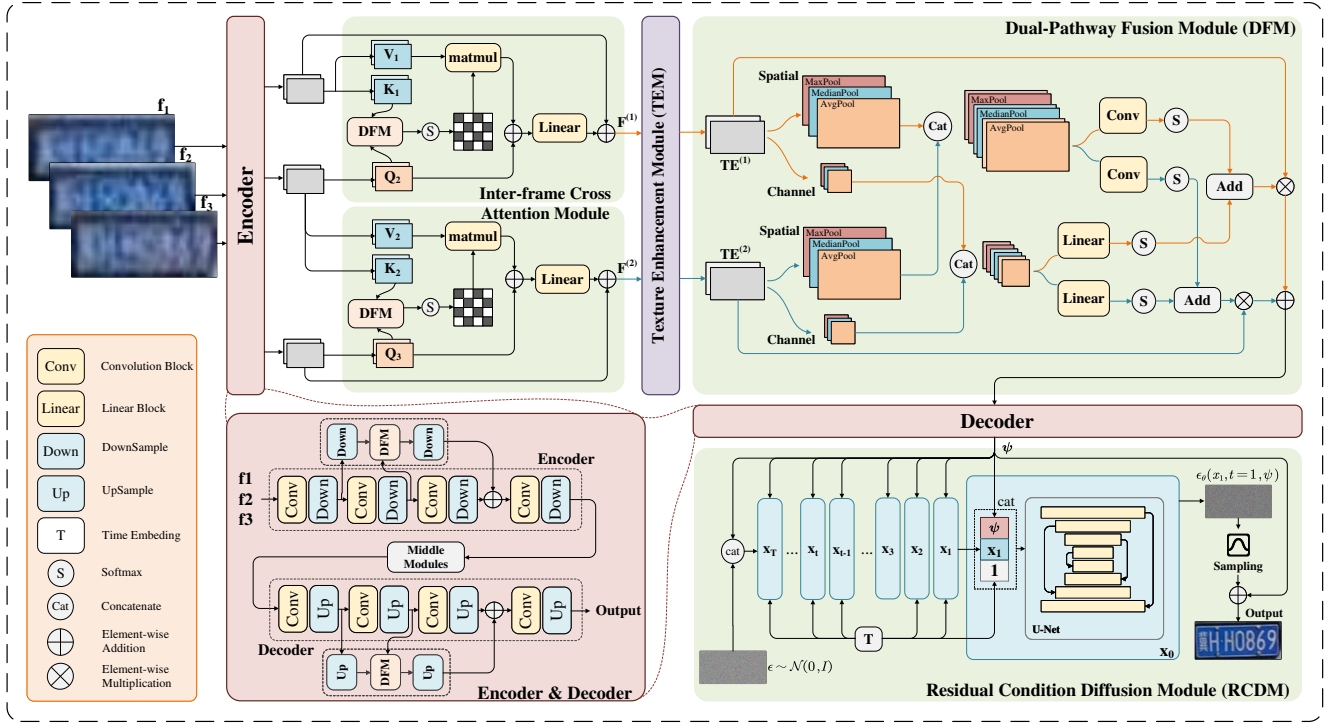


Figure 2. Overview of the proposed LP-Diff. It contains Inter-frame Cross Attention Module (ICAM), Texture Enhancement Module (TEM), Dual-Pathway Fusion Module (DFM) and Residual Condition Diffusion Module (RCDM). ICAM employs two types of cross-attention mechanisms to capture inter-frame correlations, highlighting dynamic changes. TEM enhances the texture of feature maps using curvature and gradient magnitude information. DFM filters the input features along both channel and spatial dimensions. RCDM combines the Diffusion model with residual learning to output the final clear image.

high-resolution ones for model training.

The above LPIR models rely on relatively simple degradation scenarios and are insensitive to real-world degradations. The low-resolution images used for training are also artificially generated, which significantly reduces performance when applied to real-world scenarios. Our proposed dataset contains multi-frame real-world degraded LP images with real-world degradations along with corresponding high-resolution images. On this basis, we introduce the LP-Diff that accommodates multi-frame inputs, employing several novel modules to better restore complex degradations.

3. Methodology

3.1. Overview

In this section, we introduce a Diffusion-based LP image restoration model, named LP-Diff. As illustrated in Figure 2, our method begins by using an encoder to down-sample the multi-frame images, followed by the Inter-frame Cross Attention Module (ICAM), which performs temporal fusion on the feature maps of consecutive frames, capturing relational changes and temporal dependencies. Severe degra-

dation often results in a substantial loss of texture information, compromising the model’s restoration capability. To address this issue, we propose a Texture Enhancement Module (TEM) that enhances edge information by applying Sobel filtering and calculating curvature on the feature maps. To further selectively enhance and fuse features enriched with edge information from both pathways, we integrate the Dual-Pathway Fusion Module (DFM), which fuses dual-path features in both the spatial and channel dimensions. Finally, the output is processed by a decoder to customize feature amplification, providing detailed image information to the Residual Condition Diffusion Module (RCDM) to complete the diffusion process.

3.2. Encoder and Decoder

To accurately map input features into the latent space and subsequently restore non-degraded features, we designed a pair of pre-trained encoder (E) and decoder (D), positioned at the beginning and end of the architecture, as illustrated in Figure 2.

Reducing training complexity and computational cost is essential, so we constructed the encoder and decoder using CNN modules with reduced parameters. Each module

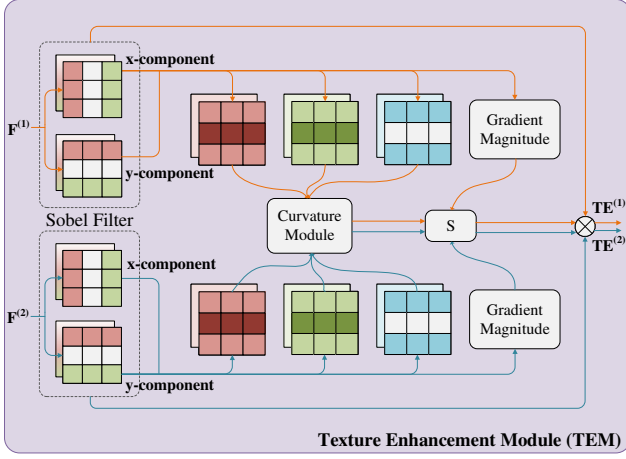


Figure 3. The structure of Texture Enhancement Module.

comprises four CNN layers combined with up-sampling or down-sampling layers. To retain more scale information during down-sampling, we incorporate a DFM, which is detailed described in Section 3.5. Specifically, the outputs from the first and second layers are fed into the DFM and subsequently combined with the output of the third layer through a residual connection. The decoder adopts a symmetrical design to maintain reconstruction accuracy during up-sampling.

We pre-train the encoder and decoder using the CCPD [40] dataset, aiming to enhance the model's learning efficiency and generalization. We set up a temporary task that uses randomly cropped images to simulate incomplete inputs, encouraging the model to fill in the missing regions. It enables the network to better reconstruct LP images by leveraging substantial prior knowledge. The encoder and decoder are trained using a combination of mean square error (MSE) and perceptual loss. For an input x_i and its corresponding ground truth y_i , the loss function is defined as follows:

$$\mathcal{L}_{\text{pretrain}} = \frac{1}{N} \sum_{i=1}^N \left[\alpha \cdot (D(E(x_i)) - y_i)^2 + \beta \cdot \|\phi_{\text{vgg}}(D(E(x_i))) - \phi_{\text{vgg}}(y_i)\|_2^2 \right], \quad (1)$$

where ϕ_{vgg} represents the outputs of the first, fifth, and tenth layers of VGG-19 [27], α and β are adjustable hyperparameters.

3.3. Inter-frame Cross Attention Module

Since consecutive frames f_1 , f_2 and f_3 contain rich temporal features that are highly correlated, they can provide important contextual information. We use the cross attention mechanism to capture the correlations between frames,

highlight dynamic changes, and suppress redundant information, leading to more accurate feature representations. After obtaining encoder outputs $E(f_1)$, $E(f_2)$ and $E(f_3)$, we designed two ICAM to leverage the contextual information from multiple frames and compensate for missing details in individual frames.

First, in the initial part, we use a linear layer to project $E(f_1)$ to K_1 and V_1 , and project $E(f_2)$ to Q_2 . We optimized the traditional cross attention module by applying the DFM module to Q_2 and K_1 . As mentioned above, there is substantial background redundancy among consecutive frames. The DFM reduces the impact of redundant pixels and channels, emphasizing the relevant features. The output is then passed through a Softmax layer to generate the weights, which are used in the dot product with V_1 . Additionally, we include two residual connections: first, the result of the weighted multiplication is combined with Q_2 , and then the output is passed through a linear layer and connected with $E(f_1)$. This retains and utilizes $E(f_1)$ and $E(f_2)$ in different forms across subsequent layers. This process can be expressed by the following equations:

$$F^{(1)} = \sigma_l \left[\sigma_s \left(\frac{DFM(Q_2 K_1)}{\sqrt{d_k}} \right) V_1 + Q_2 \right] + E(f_1). \quad (2)$$

where $\sigma_s(\cdot)$ and $\sigma_l(\cdot)$ represent Softmax and Linear layer. Similarly, in the second part, K_2 , V_2 , and Q_3 are obtained from $E(f_2)$ and $E(f_3)$ through linear transformations. The equations for the second part are as follows:

$$F^{(2)} = \sigma_l \left[\sigma_s \left(\frac{DFM(Q_3 K_2)}{\sqrt{d_k}} \right) V_2 + Q_3 \right] + E(f_3). \quad (3)$$

The difference from the first part is that we use $E(f_3)$ instead of $E(f_2)$ for the final residual connection. If we denote the distances of the capture locations for f_1 , f_2 and f_3 from the camera as d_1 , d_2 and d_3 , respectively, we assume $d_1 > d_2 > d_3$. The quality of features in $E(f_3)$ can be naturally considered more enriched compared to $E(f_2)$. Therefore, we retain more information from $E(f_3)$ in the subsequent layers.

3.4. Texture Enhancement Module

The text strokes in clear LP images typically contain delicate edges and distinct texture features, which can be utilized to distinguish text from background information. However, the text becomes bent, distorted and blurred in severely degraded LP images. Enhancing the texture features can facilitate better mitigation of these deformations and reconstruct the original text shape. Based on the above analysis, we propose the TEM, as shown in Figure 3. Initially, we set up two Sobel filter operators in x and y directions:

$$S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, S_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}. \quad (4)$$

For the outputs $F^{(1)}$ and $F^{(2)}$ from the previous module, two Sobel filters are applied to decompose them into components along the x and y directions, obtaining different high-frequency components from input path m :

$$G_d^{(m)}(i, j) = \sum_{k=-1}^1 \sum_{l=-1}^1 S_d(k, l) \cdot F^{(m)}(i + k, j + l), \quad (5)$$

where $d \in \{x, y\}$, $m \in \{1, 2\}$, and $G_d^{(m)}(i, j)$ represents the value of the output feature map at position (i, j) . Given the presence of local curvature in the strokes of letters on LP images, we apply curvature calculations [1] to these components to better describe the bending of image contours. It helps distinguish between flat regions and abrupt changes, enhancing the detection of edges and corner points. In the Curvature Module, we define two intermediate variables:

$$\mathcal{R}_x = 1 + \left(G_x^{(m)}\right)^2, \mathcal{R}_y = 1 + \left(G_y^{(m)}\right)^2. \quad (6)$$

Then we calculate the second-order partial derivatives $G_{xx}^{(m)}$, $G_{yy}^{(m)}$ and $G_{xy}^{(m)}$ in different directions, derive the curvature of the feature map, as shown in the following equations:

$$Cur^{(m)} = \frac{\mathcal{R}_y G_{xx}^{(m)} + \mathcal{R}_x G_{yy}^{(m)} - 2G_{xy}^{(m)} G_y^{(m)} G_{xy}^{(m)}}{2(\mathcal{R}_x + \mathcal{R}_y - 1)^{3/2}}, \quad (7)$$

where $m \in \{1, 2\}$. Curvature effectively describes the form and structure of changes but lacks the ability to represent the intensity of edge variations. Therefore, we introduce gradient magnitude calculation to detect and quantify the strength of these variations. The calculation formula is as follows:

$$GM^{(m)} = \sqrt{\left(G_x^{(m)}\right)^2 + \left(G_y^{(m)}\right)^2}, m \in \{1, 2\}. \quad (8)$$

After obtaining the curvature and gradient magnitude, we apply an element-wise weighting to the input features $F^{(1)}$ and $F^{(2)}$, as shown in the following equation:

$$TE^{(m)} = F^{(m)} \otimes \sigma_s \left(Cur^{(m)} + GM^{(m)} \right), m \in \{1, 2\}, \quad (9)$$

where $\sigma_s(\cdot)$ denote the Softmax layer.

3.5. Dual-Pathway Fusion Module

After obtaining the texture-enhanced features, $TE^{(1)}$ and $TE^{(2)}$ still contain channels with significant background redundancy or irrelevant pixels. We propose DFM to mitigate this issue. It uses channel attention to enhance valid channels while suppressing background and unrelated content. In addition, spatial attention is introduced to emphasize important positions within each feature map, increasing the weight of significant regions.

The DFM processes the features from previous outputs $TE^{(1)}$ and $TE^{(2)}$. Along the channel dimension, global average pooling, max pooling, and median pooling are applied to both features. Similarly, along the spatial dimension, global average pooling, max pooling, and median pooling are applied to $TE^{(1)}$ and $TE^{(2)}$. The results are then concatenated to merge similar features from both pathways, resulting in channel features and spatial features as follows:

$$\begin{aligned} F_{ch} &= \left\{ F_{ch}^{(1)}, F_{ch}^{(2)} \right\} \in \mathbb{R}^{2 \times C \times 1 \times 1}, \\ F_{sp} &= \left\{ F_{sp}^{(1)}, F_{sp}^{(2)} \right\} \in \mathbb{R}^{2 \times 1 \times H \times W}. \end{aligned} \quad (10)$$

We utilize two distinct linear modules and two separate CNN modules to generate channel features and spatial features for the two paths from F_{ch} and F_{sp} . To reassemble the features of each path, the weights of the same path are combined and multiplied by $TE^{(1)}$ and $TE^{(2)}$, respectively:

$$\begin{aligned} DF^{(m)} &= TE^{(m)} \otimes \left[\sigma_s \left(\sigma_l \left(F_{ch}^{(m)} \right) \right) \right. \\ &\quad \left. + \sigma_s \left(\sigma_c \left(F_{sp}^{(m)} \right) \right) + 1 \right], m \in \{1, 2\}, \end{aligned} \quad (11)$$

where $\sigma_s(\cdot)$, $\sigma_l(\cdot)$ and $\sigma_c(\cdot)$ denote the Softmax, Linear and Convolution layer, respectively. Finally, we apply element-wise addition to these two weighted feature maps and pass the result into the pre-trained Encoder, obtaining the aggregated embedding ψ .

3.6. Residual Condition Diffusion Module

The RCDM aims to exploit the generative capabilities of the Diffusion model to ultimately restore a clear LP image, guided by conditional inputs while employing residual learning [2]. The forward process follows a Markov chain. Given the output features ψ and the clear ground truth image f_{gt} , we take their residual as the initial image x_0 , and then gradually add Gaussian noise step by step. The forward process is represented by the following equation:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I), \quad (12)$$

where β_t is used to control the amount of noise added at each step, and I is the identity matrix. In the reverse process, it gradually recovers the residual distribution, starting from a pure Gaussian noise image at time step T :

$$p_\theta(x_{t-1} | x_t, \psi) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t, \psi), \sigma_t^2 I), \quad (13)$$

where $\mu_\theta(x_t, t, \psi)$ and σ_t^2 are the estimated mean and variance of x_{t-1} , which can be expressed as:

$$\begin{aligned} \mu_\theta(x_t, t, \psi) &= \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t, \psi) \right), \\ \sigma_t^2 &= \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t, \end{aligned} \quad (14)$$



Figure 4. Qualitative comparison with other SOTA methods, including SRCNN [5], HAT [3], Real-ESRGAN [29], ResDiff [25], and ResShift [43] on MDLP dataset. For each column, the first row’s LR consists of three degraded consecutive input frames. The text sequence above each image is the identification result from the LP recognition model. Red characters indicate recognition errors at those positions, while a red ‘-’ indicates an unrecognizable character at this position.

where $\epsilon_\theta(x_t, t, \psi)$ is the added noise during the forward process and is needed to predict by the U-net. The training objective of the RCDM is to more accurately predict the noise components in noisy samples:

$$\mathcal{L}(\theta) = \mathbb{E}_{x,t,\epsilon,\psi} \left[\|\epsilon - \epsilon_\theta(x_t, t, \psi)\|_2^2 \right]. \quad (15)$$

Since the target we set for RCDM is to restore the residual, we need to add the conditional input after the sampling. The overall process is as follows:

$$\hat{f} = \Phi_{RCDM}[\epsilon, \psi, (f_{gt} - \psi)] + \psi. \quad (16)$$

4. Experiments

4.1. Proposed MDLP Dataset

Motivations. Alongside the development of LPIR models, several LP datasets [12, 17, 19, 22, 40] have been proposed. However, these datasets primarily contain either clear or low-resolution single images, without paired data. As a result, some LPIR methods use synthetic degradation pipelines to create low-resolution images from clear LP images for model training. To address this gap, we introduce

a novel multi-frame degraded LP dataset, named MDLP. In this dataset, all the clear and degraded images are automatically collected from real-world scenarios through a well-designed pipeline. Compared to previous synthetic datasets, MDLP more accurately reflects real-world conditions and is more beneficial for model training and testing. In Section 4.6, we validate the superiority of MDLP through both quantitative and visual comparisons.

Data Collection. In practical LPIR applications, traffic surveillance systems typically capture multiple frames of blurred images. Therefore, MDLP retains all frames of the target vehicle within a specific range. We use high-speed cameras to capture fixed lanes from different angles in various road scenarios, aiming to record each frame as the vehicle moves closer or farther away. Since multiple vehicles may appear simultaneously in the lane, capturing a specific target is challenging. To address this, we use a combination of a pre-trained YOLO v5 [15] model and the DeepSort [34] algorithm during data collection to recognize vehicles, track each target, and assign a unique identifier. After collecting a large number of consecutive vehicle images, we use the YOLO v5 model to segment the LP images in each

Dataset	MDLP Dataset (Real-world)				CCPD Dataset (Synthetic)			
	PSNR↑	SSIM↑	FID↓	LPIPS↓	PSNR↑	SSIM↑	FID↓	LPIPS↓
SRCNN	14.012	0.195	248.273	0.517	12.553	0.162	251.174	0.594
HAT	14.161	0.250	229.558	0.413	12.994	0.218	235.942	0.442
Real-ESRGAN	13.927	0.369	30.975	0.176	11.741	0.297	54.058	0.246
ResDiff	12.001	0.269	35.902	0.277	10.193	0.209	59.883	0.360
ResShift	12.527	0.321	89.145	0.288	10.952	0.275	98.528	0.371
LP-Diff (ours)	14.396	0.393	21.977	0.159	13.132	0.311	46.098	0.221

Table 1. Quantitative comparison with other SOTA methods, including SRCNN [5], HAT [3], Real-ESRGAN [29], ResDiff [25], and ResShift [43] on our MDLP dataset and CCPD [40] dataset, respectively.

Model	Degradation Image	SRCNN	HAT	ESRGAN	ResDiff	ResShift	LP-Diff (ours)
NED↓	0.709	0.626	0.613	0.279	0.292	0.332	0.198
ACC↑	0.035	0.041	0.050	0.161	0.159	0.099	0.305

Table 2. Accuracy of LP text recognition comparison with other SOTA methods, including SRCNN [5], HAT [3], Real-ESRGAN [29], ResDiff [25], and ResShift [43] on our MDLP dataset.

frame.

More information about the proposed dataset MDLP can be found in the supplementary material.

4.2. Setup

Evaluation metrics. We use four evaluation metrics to evaluate the quality of generated images: PSNR, SSIM [32], FID and LPIPS [47]. PSNR and SSIM provide quantitative comparisons at the pixel and structural levels, respectively. FID and LPIPS measure the distribution distance and perceptual similarity between real and generated images. To assess the recognizability, we calculate the normalized edit distance (NED) and word accuracy (ACC) between the detected LP texts from the restored and the real images.

Implementation details. Our network uses the Adam optimizer [16], with a batch size of 20. All experiments are conducted on a GeForce RTX 3090 GPU. The total time step T of the Diffusion model is set to 1000, with 800,000 training iterations. The initial learning rate is set to 10^{-4} and decays linearly after 200,000 iterations.

4.3. Image Restoration Results

We compare our LP-Diff with other SOTA image restoration methods: SRCNN [5], HAT [3], Real-ESRGAN [29], ResDiff [25], and ResShift [43] on the MDLP dataset. As shown in Table 1, our LP-Diff outperforms other methods on both MDLP and CCPD [40] dataset. On MDLP, our LP-Diff achieves the highest PSNR and SSIM, demonstrating its ability to reconstruct clear LP images accurately. Additionally, due to its strong capability in restoring LP text, our method also achieves the best performance in FID and

LPIPS, surpassing the GAN-based method Real-ESRGAN and the Diffusion-based methods ResDiff and ResShift, revealing a higher level of realism in the restored images. Figure 4 provides a visual comparison between our LP-Diff and other SOTA models. Our model accurately reconstructs characters such as ‘X’, ‘H’, ‘M’, and numbers like ‘0’, ‘6’, and ‘9’, which are typically challenging for other models. This underscores the limitations of models designed for simple degradation in large scenes when applied to real-world LPIR. More results can be found in the supplementary material.

4.4. Text Recognition Results

The ultimate goal of the LPIR task is to improve the accuracy of automatic recognition models. Therefore, beyond image quality analysis, we further evaluate the accuracy of LP recognition. We use a pre-trained CRNN [26] model to recognize both the clear images and the images generated by each model, calculating the NED and ACC metrics for the recognized text. Additionally, we calculate the recognition results of the input degraded images to compare the enhancement achieved by each model. As shown in Table 2, our LP-Diff achieves a superior NED score compared to other models and is the only model with a value below 0.2, which indicates an LP character error rate of less than 20%. Compared to the recognition results on the original degraded images, our model improves the performance by 72.08% in NED. Compared to the next best model, NED improves by 29.03%. Our model also outperforms all other SOTA models regarding the ACC metric. Overall, LP-Diff demonstrates the best performance, highlighting its strong capability in restoring LP images.

Model	$E\&D.$	$IC.$	$TE.$	$DF.$	PSNR $\uparrow$	NED $\downarrow$
exp1		✓	✓	✓	13.757	0.231
exp2	✓		✓	✓	12.938	0.256
exp3	✓	✓		✓	12.769	0.264
exp4	✓	✓	✓		13.163	0.289
LP-Diff	✓	✓	✓	✓	14.396	0.198

Table 3. Quantitative results of ablation study. $E\&D.$, $IC.$, $TE.$ and $DF.$ denote the pre-trained Encoder and Decoder, Inter-frame Cross Attention Module, Texture Enhancement Module and Dual-Pathway Fusion Module, respectively.

4.5. Ablation Study

To evaluate the effectiveness of each proposed module, we conduct a series of ablation studies, as shown in Table 3. Without the pre-trained encoder and decoder, the reconstruction of some characters is inaccurate, and all performance metrics decline, indicating their critical importance for effective feature extraction and reconstruction. After removing ICAM, the overall metrics dropped significantly. There are also color differences between the generated and original images, which indicates the importance of this module in capturing temporal changes and maintaining color consistency. The omission of TEM leads to a notable deterioration in the NED metric, with the boundaries of the letters on the LP images becoming blurred. They confirm that the TEM enhances the model’s focus on the texture details for each character. Since the DFM is applied in the encoder, decoder, and ICAM, disabling it results in varying degrees of accuracy loss across all these modules, ultimately leading to a decrease in overall metrics. Additionally, the low utilization of output features leads to a similar functional degradation observed in exp3.

4.6. Necessity of MDLP

Since we are the first to propose a paired multi-frame degraded LP dataset, to demonstrate its superiority, we follow previous methods by applying a degradation model [22] to the CCPD [40] dataset, then obtain synthetic degraded LP images. In Table 4, we quantitatively evaluate these two types of images using frequency domain analysis, SSIM, and a blur detector. Real-world degraded images exhibit more natural degradation, with smoother and more consistent noise and blur distribution. We then perform a T-test on the above metrics, which confirms significant differences between them across all three metrics.

We further validate that MDLP enables the model to learn real-world degradation patterns better, improving its performance in practical applications. The quantitative results, summarized in Table 1 “CCPD Dataset (Synthetic)” part, clearly indicate that models trained on synthetic data are ineffective at mitigating real-world LP degradation. The



Figure 5. Comparison of results from training LP-Diff using the CCPD [40] dataset (synthetic) and our MDLP (real-world) dataset.

Dataset	f_{mean}	f_{std}	D_{blur}	SSIM
CCPD	1892.593	774.688	1348.757	0.236
MDLP	954.517	263.879	11.040	0.236

Table 4. Quantitative comparison between CCPD [40] dataset (synthetic) and MDLP (real-world) dataset. f_{mean} , f_{std} and D_{blur} represent frequency spectra mean, standard deviation and blur detector, respectively.

performance of all models declines to varying degrees when compared to their performance on MDLP data. In Figure 5, we compare the results of training LP-Diff using synthetic data versus real-world data, showing that the MDLP is more effective for training models aimed at real-world LPIR tasks.

5. Conclusion and Future Works

This paper focuses on the restoration of real-world degraded LP images. We introduce the first large-scale multi-frame degraded LP dataset, MDLP, which contains 10,245 groups of real-world severely degraded LP images. Additionally, considering the current lack of LPIR methods based on real-world data, we propose a Diffusion-based end-to-end LPIR model, LP-Diff. Experimental results show that our model performs favorably compared to the current SOTA image restoration algorithms. In the future, we intend to enhance our dataset with LP images from a broader range of countries and regions to increase its diversity. For the model, we intend to incorporate additional modules to improve the restoration accuracy of spatially complex characters. Moreover, since our model uses a Diffusion model, the inference speed is relatively slow; therefore, we will explore methods to accelerate the sampling process in future work.

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